

JOHN CARL RAMIREZ

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EDUCATION

Bachelor of Science in Computer Science

2023 - Present

University of Science and Technology of Southern Philippines, Cagayan de Oro
CGPA 1.5632 (1.0 highest), Dean's List

Elementary to Senior High School

2011 - 2023

Prophet's Pen Academy, Gitagum, Misamis Oriental
Valedictorian Grades 7-11; School President and Salutatorian in Grade 12

SKILLS AND INTERESTS

Full-Stack: React, NextJS, NodeJS, TypeScript, Django, FastAPI, Supabase, PostgreSQL, Redis, Tailwind CSS, Stripe

Mobile and Embedded: React Native, Expo, SQLite, Drizzle ORM, ESP32, PlatformIO

Testing and CI/CD: Playwright, pytest, Vitest, Jest, Maestro, Docker, GitHub Actions

Programming: Python, Java, JavaScript, TypeScript, C++, SQL, HTML/CSS

Design: Figma, Claude Design, Adobe Illustrator, Photoshop, Blender, WCAG accessibility

Software: Claude Code, GitHub, Vercel, Linux, Notion, n8n, Zapier, Premiere Pro, After Effects

Languages: English (Fluent), Bisaya (Native), Tagalog

Interests: Web and mobile development, backend systems, IoT, machine learning, data science, research, design, content creation

PROJECTS

TrailVenture, a Tour Package Booking Platform

August 2025 - November 2025

Solo Project, Full-Stack Developer

- Django REST API behind a NextJS frontend: search, package tiers, itineraries, Stripe checkout
- Server-side pricing in integer centavos behind a 30-minute hold; idempotent Stripe sessions and a signature-verified, replay-safe webhook
- Redis caching under versioned keys; 290 CI tests against a live database, Redis, and mail server; Lighthouse 100 for accessibility and SEO; live at trailventure.jcrdev.me

Steady, a Student Guidance and Counselling Platform

November 2025 - February 2026

Lead Developer, Major Project as part of curriculum

- Led a team of five building separate student, counsellor, and admin areas in NextJS and Supabase
- Access control in the database, not the interface: PostgreSQL row-level security, guard triggers, and trigger-created notifications delivered live over websockets
- 29-check database security suite and 73 CI unit tests; demo at steady-system.jcrdev.me

Agriova, an Offline-First Farm Ledger App

2026

Solo Project, Mobile Developer

- Offline-first expense, harvest, and sales tracking for Filipino smallholder farmers; Android and iOS from one TypeScript codebase
- Money as integer centavos, every write in one transactional layer with a sync outbox; 56dp touch targets and full Bisaya and English for field use
- In-app Gemini assistant answering from the farmer's own records; Jest and Maestro tests in CI

Post-Disaster Road Restoration using Steiner Tree Approximation Algorithms

August 2025 - Present

Corresponding Author, Major Project as part of curriculum

- Novel approaches to optimizing post-disaster road restoration with Steiner tree approximation algorithms
- Held the 2-approximation bound across 30 random graphs, twelve Istanbul benchmarks, and Cagayan de Oro road data, at near-optimal ratios and fast runtime
- Groundwork for my thesis on learning-augmented online road restoration: reinforcement learning as predictor, genetic algorithm as planner

Lead Video Editor for the University-Wide Intramurals

January 2026

University Student Government, USTP-CDO

- Edited 10 event videos in Premiere Pro, After Effects, Photoshop, and Blender, including the Drag Queen promo and short-form content; contributed to 100k+ cumulative social media engagements

RESEARCH PUBLICATION

John Carl Ramirez, Gerlie Campion, Kathleen Grace Gultiano, Junar Landicho "*Performance Analysis and Application of KMB 2-Approximation Algorithm in Finding Optimal Routes for Post-Disaster Road Network Restoration*"

SSRN preprint: <http://dx.doi.org/10.2139/ssrn.5277559>

May 2025

WORK EXPERIENCE

Enduro Group Business Consulting and Services

December 2025 - August 2026

Lead Designer and Branding Manager, Dallas, Texas

- Authored the firm's 31-page brand guidelines (v2.0) in Figma and Claude Design: wordmark, clearspace, colour, type, and application rules
- Identity systems and responsive web pages for the Al-Bab Initiative and Xplore Land & Sea, to WCAG standards
- Design operations workflow in Claude Code and Design, shipping each brand as a version-controlled GitHub repository

Fresco Greenovations Inc.

2026

Solo Developer, Internship, Cagayan de Oro

- Sole developer of Fresco Grow Lab, an IoT telemetry system for grow-bag experiments
- ESP32 boards in C++ with PlatformIO logging temperature at four depths and tipping-bucket rainfall, flashed and configured from the browser
- NextJS dashboard turning readings into watering, thermal, and rainfall analytics

STUDENT LEADERSHIP

Computer Science Student Society (CS3)

November 2025 - Present

CS3 Core Team Lead

- Lead a selected group of CS students on research and software solutions; building a unified website for USTP-CDO's Department of Computer Science and CS3, and restructuring the team

EXTRA-CURRICULAR

- Top 20 semifinalist, **Innovation Cup Mindanao 2026** June 2026
- Completed **Wadhvani Foundation's Ignite Philippines** entrepreneurship coursework May 2026
- Participated in DICT's **AI.Deas Region X**, Cagayan de Oro City September 2025
- Level 2, **11th TOPCIT** (Test of Practical Competency in IT) November 2024
- Completed **freeCodeCamp's Responsive Web Design** certification August 2024
- 1st Place in USTP-CDO, **Google Developer Student Clubs APAC Solution Challenge**, competed at Asia-regional level March 2024

- Member, University Student Government (Video Editing and Multimedia Head), 4-H Club USTP-CDO, and CS3 2024 - Present

DECLARATION

I hereby declare that all the details furnished above are true to the best of my knowledge and belief.